

JavaScript Practical Programs – Flow, Logic, and Methods

1. Change Background Color on Button Click

- Flow: Button click → onclick event → JavaScript function executes → document.body.style.backgroundColor changes.
- Methods: onclick (handles click), document.body (accesses page body), style.backgroundColor (changes background color).

2. Recognize Mouse Events

- Logic: Different mouse actions trigger different events.
- Methods: onclick, ondblclick, onmouseover, onmouseout, onmousedown, onmouseup.

3. Recognize Keyboard Events

- Flow: User presses/releases a key → event fires → display key information.
- Logic: Detect keyboard activity.
- Methods: onkeydown, onkeyup, onkeypress (legacy), event.key, event.keyCode (legacy).

4. Recognize Form Events

- Flow: User interacts with form → focus/change/submit/reset events occur.
- Logic: Respond to user actions and validate before submission.
- Methods: onfocus, onblur, onchange, onsubmit, onreset, preventDefault().

5. Change HTML Element Properties

- Flow: Select HTML element → modify content or style.
- Logic: Dynamically update webpage.
- Methods: getElementById(), querySelector(), innerHTML, textContent, style.color, style.fontSize, style.display.

6. Student Registration Form Validation

- Flow: Read input values → validate → show errors or submit.
- Logic: Prevent invalid data.
- Methods: value, trim(), match()/Regex, isNaN(), alert(), preventDefault().

7. CRUD Operations on Array

- Flow: Create → Read → Update → Delete → Refresh display.
- Logic: Store and manipulate data in an array.
- Methods: push(), splice(), indexOf(), forEach(), join().

8. Mouse Event Practical

- Hover heading → onmouseover changes text to yellow; onmouseout changes back to black.

- Click 'Find Time' → onclick displays current date/time using new Date() and toLocaleString().
- Click Red/Green buttons → change background using style.backgroundColor.

9. Keyboard Event Practical

- Flow: onkeydown detects pressed key. If key is a,e,i,o,u then display 'Vowel pressed'.
- Flow: onkeyup changes page background to blue.
- Methods: onkeydown, onkeyup, event.key, toLowerCase(), includes(), style.backgroundColor.

Common Methods and Their Purpose

Method / Event	Purpose
onclick	Detect button click
onmouseover	Mouse enters element
onmouseout	Mouse leaves element
onkeydown	Key pressed
onkeyup	Key released
onfocus	Input selected
onblur	Input loses focus
onchange	Input value changed
onsubmit	Form submitted
getElementById()	Select element by ID
querySelector()	Select first matching element
innerHTML	Change HTML content
textContent	Change text only
style	Modify CSS properties
value	Read input value
preventDefault()	Stop default action
push()	Add array item
splice()	Insert/Delete/Update array
forEach()	Loop through array
new Date()	Current date and time